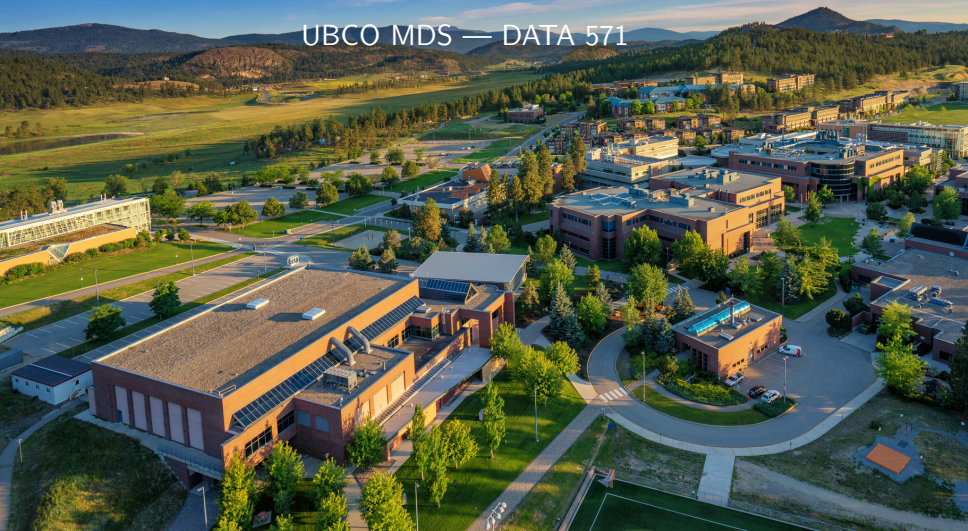


Association Rules

UBCO MDS — DATA 571





- ▶ Association rule mining is primarily used to discover associated items in large databases.
- ▶ We are looking for simple rules (and probabilities associated with those rules) that tell us “if this, then that”
- ▶ Combinatorially speaking, this is a challenging problem. But conceptually it is pretty straightforward. Let’s learn through the most common ARL example — purchasing groceries.

Grocery Example



- ▶ We have 30 days of point-of-sale transaction data from a grocery store. Think of your receipt as a single observation in the data.
- ▶ We can formulate that receipt (observation) into a binary vector of all categories of items that the grocery store sells.
- ▶ For example, if you bought milk and eggs but not cookies or bread then

Observation	Milk	Cookies	Bread	Eggs	...
Receipt 1	.	.	.	.	...
Receipt 2 (yours)	1	0	0	1	...
Receipt 3	.	.	.	.	...
...	.	.	.	.	...



- ▶ For the actual data, we have 9835 transactions (receipts, observations, etc) and have aggregated all of the purchases into 169 categories (also called items).
- ▶ So, our hope is to figure out which sets of items (itemsets) are co-purchased
- ▶ This will help us generate rules like: “if pasta and chicken are purchased, then we also expect cheese to be purchased”
- ▶ From a marketing standpoint, the utility of such information should be obvious. Cheese companies may want to include coupons in the pasta aisle in the hopes that the person will select their cheese over a competitor, for example.



- ▶ Naively, we want to consider all possible itemsets for an association with all other possible itemsets.
- ▶ This reasoning gets out of hand quickly!
- ▶ Each itemset can either have, or not have, each individual item. So how many unique itemsets are there? Board.
- ▶ How many possible association rules exist? Board.

- ▶ Notation wise, let's let each transaction be

$$t_i = \{x_{i1}, x_{i2}, \dots, x_{id}\}$$

where each x_{ij} equals 0 or 1 indicating whether item j was purchased. Suppose d items and n transactions.

- ▶ We seek rules that tell us that itemset $Y \rightarrow Z$ with some measurement(s) of that discovered relationship.
- ▶ We'll say that itemset $Y = \{x_k = 1, x_l = 1, \dots\}$ is equivalent to the set $\{k, l, \dots\}$, meaning that the itemset consists of items k , l , and...

- ▶ **Support** is the fraction of transactions that a particular itemset, say $M = \{x_k = 1, x_l = 1, \dots\}$, appears

$$\text{Supp}(M) = \frac{\sum_{i=1}^n I(M \in t_i)}{n}$$

- ▶ So, if the itemset of interest only includes one item $M = \{x_j = 1\}$, then the support is $\frac{\sum_{i=1}^n x_{ij}}{n}$, or simply the proportion time that item j was purchased.
- ▶ So support indicates the proportion of time all items in the itemset are purchased together.

- ▶ We're seeking measures between itemsets, so
- ▶ For itemsets Y and Z , **confidence** for the relationship $Y \rightarrow Z$ is the proportion of transactions that contain both itemsets Y and Z divided by those that only contain Y .

$$\text{Conf}(Y, Z) = \frac{\text{Supp}(Y \cup Z)}{\text{Supp}(Y)} = \frac{\sum_{i=1}^n I(Y \in t_i; Z \in t_i)}{\sum_{i=1}^n I(Y \in t_i)}$$

- ▶ AKA, how likely itemset Z is purchased given that itemset Y was purchased (which implicitly controls for the popularity of Y).
- ▶ See board for discussion on confidence...

- **Lift** adjusts confidence by altering the denominator to be the expected $\text{Supp}(Y, Z)$ assuming that Y and Z can be considered independent itemsets

$$\text{Lift}(Y, Z) = \frac{\text{Supp}(Y \cup Z)}{\text{Supp}(Y) \times \text{Supp}(Z)} = \frac{n \sum_{i=1}^n I(Y \in t_i; Z \in t_i)}{\sum_{i=1}^n I(Y \in t_i) \times \sum_{i=1}^n I(Z \in t_i)}$$

- AKA, a measure of how likely itemset Z is purchased given that itemset Y was purchased, controlling for the popularity of Z (and Y).
- $\text{Lift}(Y, Z) = 1$ suggests independence. $\text{Lift}(Y, Z) > 1$ suggests positive relationship. $\text{Lift}(Y, Z) < 1$ suggests negative relationship.



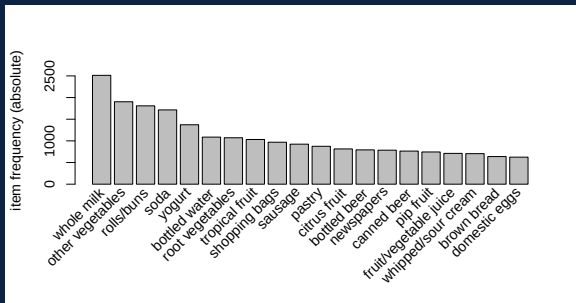
- ▶ Now, back to the problem at hand.
- ▶ We can't calculate these measures for all possible $Y \rightarrow Z$ itemset comparisons
- ▶ But, especially in cases like the grocery example where most transactions will only include a small subset of items, we can simplify greatly...
- ▶ First, we consider only investigating rules that meet minimum thresholds of support and confidence.

- ▶ Next, we recognize that the support of any itemset cannot exceed the support of its subsets.
- ▶ So for example, if $Y = \{j\}$ and $Z = \{j, k, \dots\}$ then $\text{Supp}(Y) \geq \text{Supp}(Z)$
- ▶ And so, when a relatively simple itemset Y is found to have support below the threshold, we don't have to consider any larger itemsets that contain Y .
- ▶ See board



- ▶ Furthermore, some softwares only consider single items for the right hand side of the relationship (Z).
- ▶ Let's take a look at the grocery data...

► Quick exploration of the data



AR on Groceries



```
> summary(arun)
set of 410 rules
```

```
rule length distribution (lhs + rhs):sizes
```

```
  3   4   5   6
29 229 140  12
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
3.000	4.000	4.000	4.329	5.000	6.000

```
summary of quality measures:
```

support	confidence	lift	count
Min. :0.001017	Min. :0.8000	Min. : 3.131	Min. :10.00
1st Qu.:0.001017	1st Qu.:0.8333	1st Qu.: 3.312	1st Qu.:10.00
Median :0.001220	Median :0.8462	Median : 3.588	Median :12.00
Mean :0.001247	Mean :0.8663	Mean : 3.951	Mean :12.27
3rd Qu.:0.001322	3rd Qu.:0.9091	3rd Qu.: 4.341	3rd Qu.:13.00
Max. :0.003152	Max. :1.0000	Max. :11.235	Max. :31.00

```
mining info:
```

data	ntransactions	support	confidence
Groceries	9835	0.001	0.8



	lhs	rhs	support	confidence	lift	count
[1]	{citrus fruit, tropical fruit, root vegetables, whole milk}	=> {other vegetables}	0.003152008	0.8857143	4.577509	31
[2]	{other vegetables, curd, domestic eggs}	=> {whole milk}	0.002846975	0.8235294	3.223005	28
[3]	{hamburger meat, curd}	=> {whole milk}	0.002541942	0.8064516	3.156169	25
[4]	{herbs, rolls/buns}	=> {whole milk}	0.002440264	0.8000000	3.130919	24
[5]	{tropical fruit, herbs}	=> {whole milk}	0.002338587	0.8214286	3.214783	23

```
> inspect(sort(arun, by="confidence")[1:5])
```

	lhs	rhs	support	confidence	lift	count
[1]	{rice, sugar}	=> {whole milk}	0.001220132	1 3.913649	12	
[2]	{canned fish, hygiene articles}	=> {whole milk}	0.001118454	1 3.913649	11	
[3]	{root vegetables, butter, rice}	=> {whole milk}	0.001016777	1 3.913649	10	
[4]	{root vegetables, whipped/sour cream, flour}	=> {whole milk}	0.001728521	1 3.913649	17	
[5]	{butter, soft cheese, domestic eggs}	=> {whole milk}	0.001016777	1 3.913649	10	


```
> inspect(sort(arun, by="lift")[1:5])
```

	lhs	rhs	support	confidence	lift
[1]	{liquor, red/blush wine}	=> {bottled beer}	0.001931876	0.9047619	11.235269
[2]	{citrus fruit, other vegetables, soda, fruit/vegetable juice}	=> {root vegetables}	0.001016777	0.9090909	8.340400
[3]	{tropical fruit, other vegetables, whole milk, yogurt, oil}	=> {root vegetables}	0.001016777	0.9090909	8.340400
[4]	{citrus fruit, grapes, fruit/vegetable juice}	=> {tropical fruit}	0.001118454	0.8461538	8.063879
[5]	{other vegetables, whole milk, yogurt, rice}	=> {root vegetables}	0.001321810	0.8666667	7.951182



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